VOIDSCOPE · FEASIBILITY ASSESSMENT

VoidScope feasibility assessment

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First issue (kit 1.9.1 upgrade)

VoidScope feasibility assessment

VoidScope option A is feasible on paper as a hand-built open design with commodity parts and one outside lathe job; the weakest points are the bought camera's real power and sealing, the bite valve and the survivor-facing water line, all of which only TRL 4 tests and a medical lead can settle. Method: STANDARDS section 23.

> **Safety:** VoidScope is rescue equipment that may supply water and air to a trapped person and carries a lithium battery. This document is an open engineering reference, not certified rescue equipment, not a medical device and not legal advice.

Table 1. Feasibility by aspect (option A, the design of record).

Aspect	Assessment	Rating	Risks and notes	Evidence
Design	A 4.5 m rigid rod passes a 51 mm hole, buckles at 940 N and steers its tip 90 degrees each way; all calculations meet their targets on paper	High	Estimates only; wire breaking load, camera power and cable stiffness are assumed	VDS-STM-001, VDS-CAL-001, D3 model (cad/fidelity.json)
Materials	6061-T6 tube, stainless pins and wire, PA12 printed links, silicone sheath, PTFE guide; all commodity	High	Galvanic attack in a wet shift (STM S16); PA12 links are the new weak point (printed, drop-test at TRL 4)	VDS-STM-001 S16, S6
Manufacturing and assembly	Tube cut with hand tools; turned head, tip housing and tail cap need a lathe (outside shop allowance USD 240); links printed in nylon	Medium	The nose and collar need an offset four-jaw set-up; tolerances on the telescoping fit (0.11 mm a side) are tight	VDS-BLD-001, VDS-DDR-002
Power and controls	One 12.8 V 6 Ah LiFePO4 pack gives 6.2 h of video at 20 C and 4.4 h at -10 C; blower speed set by a knob	High	Cold capacity assumed 70 per cent; charging only above 0 C	VDS-CAL-001 F, STM S15, S17
Water and air delivery	Gravity drip capped at 5 mL/min by a capillary; air through the rod bore at 0.9 kPa at most	Medium	The drip ceiling has a 12 per cent margin at 2 m and 40 C with an oversize bore; bite valve cracking pressure unknown; aspiration is a clinical risk	STM S7, S9, S11, S18; RSK H1
Thermal	Tip housing 44 C at 35 C ambient (54 C at 45 C) after the fix (black anodised, LEDs dimmed)	Medium	Near the limit at 45 C ambient; the real camera power is unknown	VDS-STM-001 S14, VDS-DDR-006 F1
Supply of parts	Every BOM line is a catalogue part or stock; the camera head, monitor and blower are the single-source risks	Medium	Prices are first estimates; no supplier has been asked to quote	bom/bom.csv
Cost	USD 1,780 against a USD 2,000 value-engineering target	High	Control target, not a limit; B and C probe parts unpriced	bom/bom.csv, VDS-OPT-001
Use and training	Two people, 2.2 min estimated setup; written drill card	Medium	Only timed drills settle it; needs a trained team	VDS-BLD-001 5a

Ratings: High, Medium, Low. No aspect is rated Low.